

xMCF

ISO/PAS 8329

Description of mechanical connections
and joints in structural systems

ISO/PAS 8329 xMCF WG meeting
2025-12-10

Meeting results are
added in green font.
Additions are
inserted in light blue.

Agenda

- 1 Recent progress & next steps with respect to [#105](#) and [#61](#)
- 2 ~~xMCF-related findings in PTC-Creo~~ — Unfortunately, PTC couldn't attend. ⇒ Item postponed.
- 3 Planning Kickoff 2026
- 4 Arranging Ulrich Fox's succession

GitHub Issues — Top 6

<https://dud-poll.inf.tu-dresden.de/xMCF/> — as of 2025-11-12, 11:00 CET

Top 	Med 	#	Title	Volunteers
6	0	105	Support connection 0/1/2d to be reused (e.g. multiple occurrences)	CF, UF, NE, SH
4	2	61	xMCF shall Support Catalogs, e.g. of Standard Parts / Joining Elements	CF, UF, NE, JB
4	2	92	Setup a Document with Recommendations / Best Practices of xMCF Usage	CF, IR
4	2	95	Add persistent identifier to connection group	CF, UF, GH, NE, SH, TR
3	2	94	Add instance identifier to a connector	CF, UF, GH, NE, SH, TR
3	1	59	Mapping between xMCF and other established systems	CF, TR

Comments & Hints:

- Public example CAD files are attached to [#104](#), which could be useful for [#92](#).



Short recap. Where are we today?

- In the last meeting, we made initial, fundamental decisions regarding the direction to take wrt. issues [#105](#) and [#61](#).
- Today we will continue our work at this point.



Organizational Action Items: Version Code and GitHub

- **Versioning:** Since we plan to extend xMCF 3.1.1 by solving issues [#105](#) and [#61](#) in an upward compatible manner, and will do more than just “some bug fixing”, I propose increasing the xMCF version code to 3.2. (This follows the concept of “[Semantic Versioning](#)”.)
Confirmed.
- **Content:** Do we want to publish the new version as soon as issues [#105](#) and [#61](#) are finished, or do we want to collect more issues, first?
We want to include the issues [#94](#) and [#95](#), too.
- **Milestone:** I would then rename the GitHub milestone “[Release after V3.1.1](#)” to “Release 3.2” and gather the respective issues, there.
The proposal has been confirmed and already implemented.
Suggestions for a planned release date? ... summer 2026. This would require xMCF 3.2 to be submitted to ISO by April/May 2026.
- **Branching:** To keep the changes separate from the last released version (branch “[master](#)”), I would create a new branch called ‘*issues_61_94_95_105*’ incl. directories for *examples*, *resources* and *test_suite* beneath new directory V3.2. Here we can work together on the new results.
The proposal has been confirmed and already implemented.

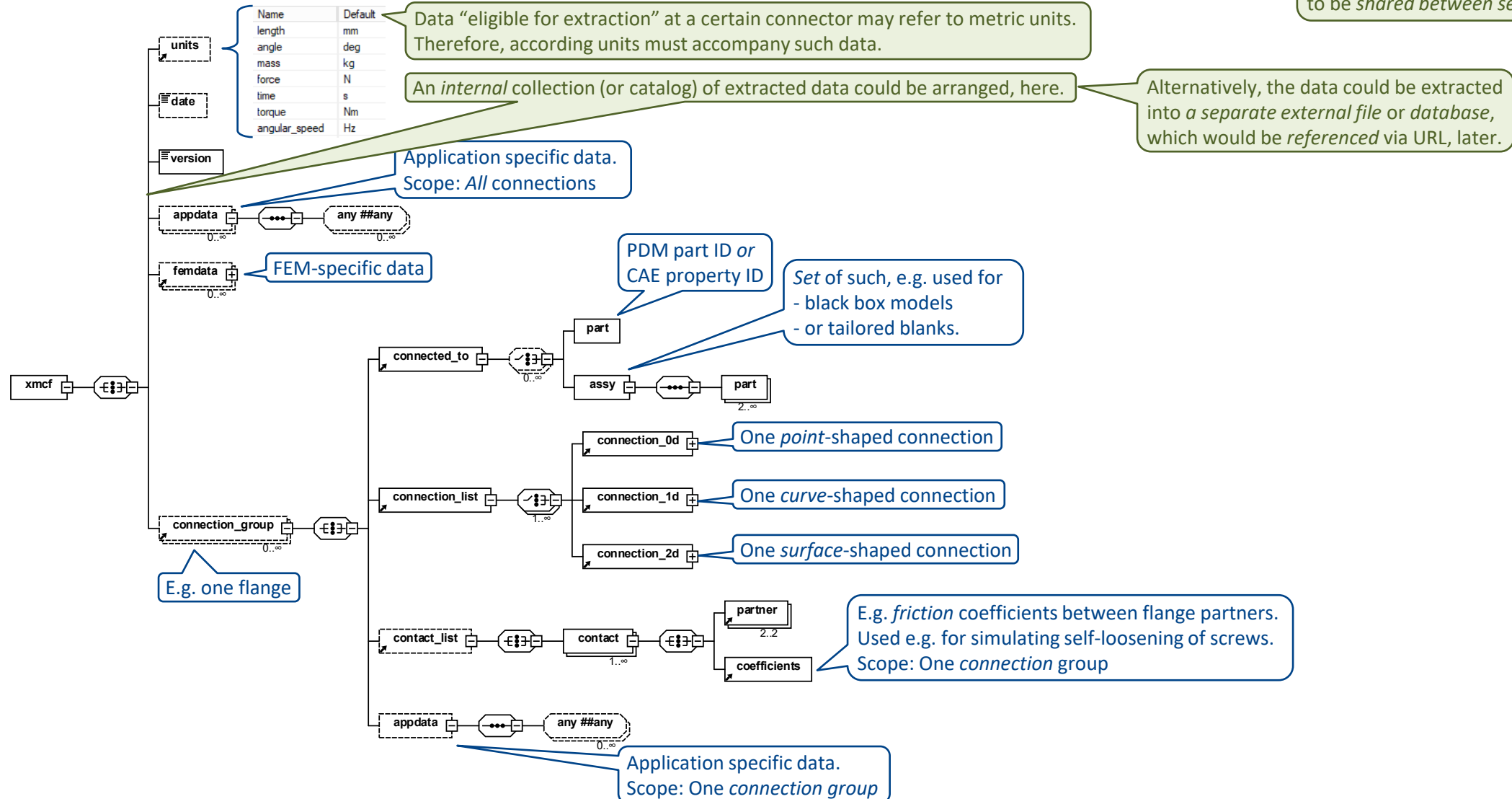


Contentual Action Items:

- Standard document (Word file)
 - Description of overall concept & objectives, intended application & motivation etc.
 - Description of semantics for internal catalog
 - Description of semantics for reference to a catalog item
 - Definition of example use cases and in-document XML-examples
- XML & XSD — CADENCE/BETA CAE will contribute to this.
 - Sketching XML example file(s), incl.
 - precedence rule and dropped mandatory attribute,
 - internal and external file-based catalog
 - Description of syntax for catalogs
 - Description of syntax for reference to a catalog item:
 - internal, external file-based, external database-based
 - Drafting XML schema (XSD) and test cases / test example files

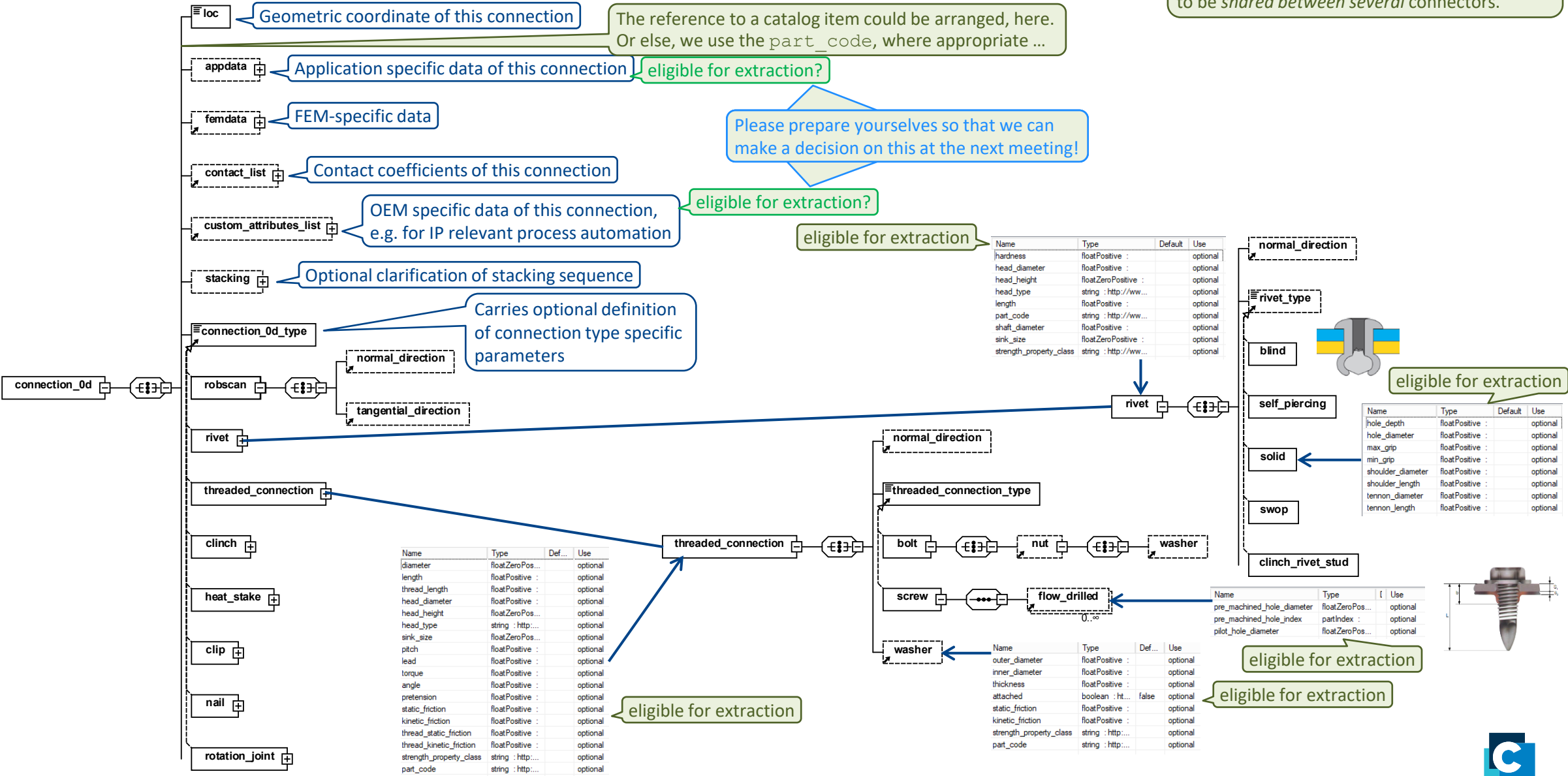


Structure of χ MCF 3.1.1 XSD – High Level



Structure of xMCF 3.1.1 XSD – Point Connections

“eligible for extraction” to be understood as:
Data can be *taken out of one* specific connector,
to be *shared between several* connectors.



Questions & Considerations of the 2025-11-26 Meeting

1. Q: What happens, if a connector contains both, a reference to extracted data (catalog) *and* (some of) such data, internally?
Which data takes precedence?
A: “Internal” data shall take precedence.
2. Q: May a connector omit *mandatory* attributes if they can be determined via the reference to a catalog entry?
A: Yes, but schema validation must take this into account.
3. Q: Should a catalog entry be allowed to refer to other entries, maybe even in another catalog?
Could be used to enable a *step-by-step completion*.
A: Not allowed, for internal catalogs, for the moment.
Format of database-catalogs is not subject of the standard, therefore no statement given here.
4. Q: Should we allow both, *internal* & *external* catalogs?
Even in combination?
A: Yes.

- Tools need to store, where data “eligible for extraction” originally was read from (individual connector, internal or external catalog).
- For new connectors, users must be able to specify and for existing ones, users must be able to modify this. In both cases, incl. the URL of the external catalog.
- Q: Should the system allow the modification of data originating from an external catalog?
A: Yes, but this creates internal data.
- Q: When writing an χ MCF file:
Should the system emit a warning if data in an *external* catalog has changed, meanwhile?
Or, if the external catalog can no longer be found. (To avoid lost or inconsistent data, in the end.)
A: References to external databases must consider life cycle (e.g., release codes) of the items contained therein.
However, we can leave this responsibility to the process owner. (Can be discussed further, later.)



Proposed wording for results of our previous meeting

- We will focus entirely on `connection_0d` and leave `connection_1d` and `connection_2d` aside for the moment.
However, we will take care not to impair any future expansion.
- We will allow both, *internal* & *external* catalogs, even in combination. — Where:
- An *internal* catalog resides *within* the specific χ MCF file at root level, as a sibling element of, for example, `<units/>`.
- An *external* catalog is located *outside* the specific χ MCF file. It can be a simple file, a collection of data records in a database or something else.
The standard should be formulated agnostically in this regard, with any reference to items in the external catalog described by a URL.

Proposal for continuation:

- If the *external* catalog is a simple file, its syntax should follow the already defined χ MCF syntax as closely as possible.
Future extensions to the basic syntax should, if possible, automatically yield the syntax extension for the external file.

Confirmation? ...

Proposed wording for results of our previous meeting

- If a connector in a χ MCF file contains both,
 - a reference to catalog data,
 - and (some of) such data internally in its definition,the internal data shall take precedence.
- A connector may omit otherwise mandatory attributes if these can be determined via a reference to a catalog entry. XML schema validation must take this into account.
- Note (recursive references): A catalog entry *in χ MCF format* (either internal or external file-based) may *not* (for the time being) refer to other catalog entries itself. The format of *database-based* catalog entries is not subject to the standard. Therefore, no statement is made here.

At some point in the future, however, this feature could still prove useful for enabling the step-by-step completion of connector definitions.



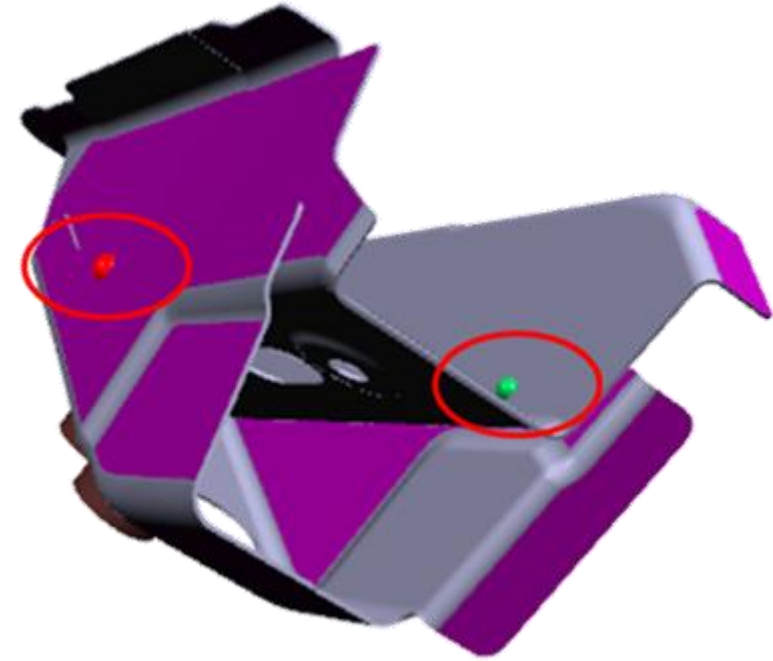
Other ISO Standards

[#61](#) suggests to analyze existing standards in this context, such as

- [ISO 13584-511:2006 — Industrial automation systems and integration — Parts library — Part 511: Mechanical systems and components for general use — Reference dictionary for fasteners](#)
- [ISO 1891:2009 — Fasteners — Terminology](#) — a copy was provided to the WG — for internal use, only.
- Anymore?
- Organizational question: How can we get copies of such standards?
⇒ Homework for all: Please see if you can find some useful information “at home”!

Guillaume's questions of the 2025-11-26 meeting

- Catalogue, as identified using [VDA231-200](#) used at BMW: standard type (DIN EN, GS, SAE, ...), standard number, legal owner (especially important for internal company standards), issue date → [May be mapped to components of the URL?](#)
- `part_code`: At BMW, the welding points (spotwelds) are represented by a small sphere, which has a part number (see graphic) (not for logistic purposes, since it cannot be ordered or manufactured) → [We should transport this code.](#)
[The attribute name remains to be discussed, though.](#)
- Potential attributes dedicated to each *individual* usage of a fastener: location/orientation, *instance number* (see [#94](#)), *tightening torque*. → [To be considered!](#)
- Difference between `connection_[012]d.label` and `part_code`? → See issue [#104](#).
→ [Maybe, formulation in standard document needs improvement.](#)
[Please leave a hint in \[#104\]\(#\), in this case.](#)



Ulrich Fox's Retirement

Today is the last time Ulrich Fox attends our working group, as he is retiring at the end of the year.

The working group would like to express its sincere thanks
for all the essential work he carried out over the past years!

Thank You

Ulrich Fox's succession as co-lead of ISO 8329/χMCF working group

We are still looking for a successor. — Job description:

- Facilitate monthly core team meetings
- Facilitate quarterly full forum meetings
- Contribute and co-lead project management
- Contribute to discussion and resolution of issues and requirements
- Contribute to technical documentation

Average time consumption 2h/week

- [Suggestions / Volunteers: ...](#)

Next meetings are to be scheduled ...

- Kickoff 2026: 2026-01-14, 15:00 CET — CF will send invitation.
Focus topics:
 - Recent progress & next steps with respect to
 - [#105](#) — [Support connection 0/1/2d to be reused \(e.g. multiple occurrences\)](#)
 - [#61](#) — [xMCF shall Support Catalogs, e.g. of Standard Parts / Joining Elements](#)
 - Syntax proposals & examples are very welcome!
 - Issues [#94](#) and [#95](#) are kept on the backlog, for the next meeting.

